

ANOVA: Two-Factor Without Replication

Easy, Line-by-Line Calculation

Data: Dept A = 82, 86, 79, 88; Dept B = 78, 81, 75, 84; Dept C = 85, 89, 84, 92; Dept D = 80, 83, 78, 86; Dept E = 88, 91, 87, 94.

1. Count Rows, Columns and Observations

Rows = 5 departments.

Columns = 4 machines.

Total observations = $5 \times 4 = 20$.

2. Row Means

Dept A: $(82 + 86 + 79 + 88) / 4 = 335 / 4 = 83.75$

Dept B: $(78 + 81 + 75 + 84) / 4 = 318 / 4 = 79.50$

Dept C: $(85 + 89 + 84 + 92) / 4 = 350 / 4 = 87.50$

Dept D: $(80 + 83 + 78 + 86) / 4 = 327 / 4 = 81.75$

Dept E: $(88 + 91 + 87 + 94) / 4 = 360 / 4 = 90.00$

3. Column Means

Machine 1: $(82 + 78 + 85 + 80 + 88) / 5 = 413 / 5 = 82.6$

Machine 2: $(86 + 81 + 89 + 83 + 91) / 5 = 430 / 5 = 86.0$

Machine 3: $(79 + 75 + 84 + 78 + 87) / 5 = 403 / 5 = 80.6$

Machine 4: $(88 + 84 + 92 + 86 + 94) / 5 = 444 / 5 = 88.8$

4. Grand Mean

Total of all values = 1690.

Grand Mean = $1690 / 20 = 84.5$.

5. SS Rows

Formula: $SS \text{ Rows} = \text{Number of Columns} \times \sum (\text{Row Mean} - \text{Grand Mean})^2$

Dept A: $4(83.75 - 84.5)^2 = 4(0.5625) = 2.25$

Dept B: $4(79.5 - 84.5)^2 = 4(25) = 100$

Dept C: $4(87.5 - 84.5)^2 = 4(9) = 36$

Dept D: $4(81.75 - 84.5)^2 = 4(7.5625) = 30.25$

Dept E: $4(90 - 84.5)^2 = 4(30.25) = 121$

Add: $2.25 + 100 + 36 + 30.25 + 121 = 289.5$

6. SS Columns

Formula: $SS \text{ Columns} = \text{Number of Rows} \times \sum (\text{Column Mean} - \text{Grand Mean})^2$

M1: $5(82.6 - 84.5)^2 = 5(3.61) = 18.05$

$$M2: 5(86 - 84.5)^2 = 5(2.25) = \mathbf{11.25}$$

$$M3: 5(80.6 - 84.5)^2 = 5(15.21) = \mathbf{76.05}$$

$$M4: 5(88.8 - 84.5)^2 = 5(18.49) = \mathbf{92.45}$$

$$\text{Add: } 18.05 + 11.25 + 76.05 + 92.45 = \mathbf{197.8}$$

7. SS Total

Formula: $SS\ Total = \sum(X - \text{Grand Mean})^2$

Example: $(82 - 84.5)^2 = (-2.5)^2 = 6.25$.

Do this for all 20 values and add them.

SS Total = 491

8. SS Error

Formula: $SS\ Error = SS\ Total - SS\ Rows - SS\ Columns$

$= 491 - 289.5 - 197.8$

$= 201.5 - 197.8$

= 3.7

9. Check SS

$SS\ Total = SS\ Rows + SS\ Columns + SS\ Error$

$= 289.5 + 197.8 + 3.7 = \mathbf{491\ \checkmark}$

10. Degrees of Freedom (df)

Rows $df = 5 - 1 = 4$

Columns $df = 4 - 1 = 3$

Error $df = (5 - 1)(4 - 1) = 4 \times 3 = \mathbf{12}$

Total $df = 20 - 1 = \mathbf{19}$

11. Mean Square (MS)

$MS = SS / df$

Rows $MS = 289.5 / 4 = \mathbf{72.375}$

Columns $MS = 197.8 / 3 = \mathbf{65.9333}$

Error $MS = 3.7 / 12 = \mathbf{0.308333}$

12. F Value

$F = MS\ Factor / MS\ Error$

Rows $F = 72.375 / 0.308333 = \mathbf{234.7297}$

Columns $F = 65.9333 / 0.308333 = \mathbf{213.8378}$

13. Final ANOVA Table

Source	SS	df	MS	F
Rows	289.5	4	72.375	234.7297
Columns	197.8	3	65.9333	213.8378
Error	3.7	12	0.308333	

Total	491	19		
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14. Easy Formula Sheet

Rows SS: $\text{Columns} \times \Sigma(\text{Row Mean} - \text{Grand Mean})^2$

Columns SS: $\text{Rows} \times \Sigma(\text{Column Mean} - \text{Grand Mean})^2$

Error SS: $\text{Total SS} - \text{Rows SS} - \text{Columns SS}$

Total SS: $\Sigma(X - \text{Grand Mean})^2$

MS: SS / df

F: $\text{MS Factor} / \text{MS Error}$

Rows df: $\text{Rows} - 1$ | **Columns df:** $\text{Columns} - 1$

Error df: $(\text{Rows} - 1)(\text{Columns} - 1)$ | **Total df:** $N - 1$

Learning order: Row Mean $\rightarrow$ Column Mean $\rightarrow$ Grand Mean $\rightarrow$ Rows SS $\rightarrow$ Columns SS $\rightarrow$ Total SS $\rightarrow$ Error SS $\rightarrow$ df $\rightarrow$ MS $\rightarrow$ F.